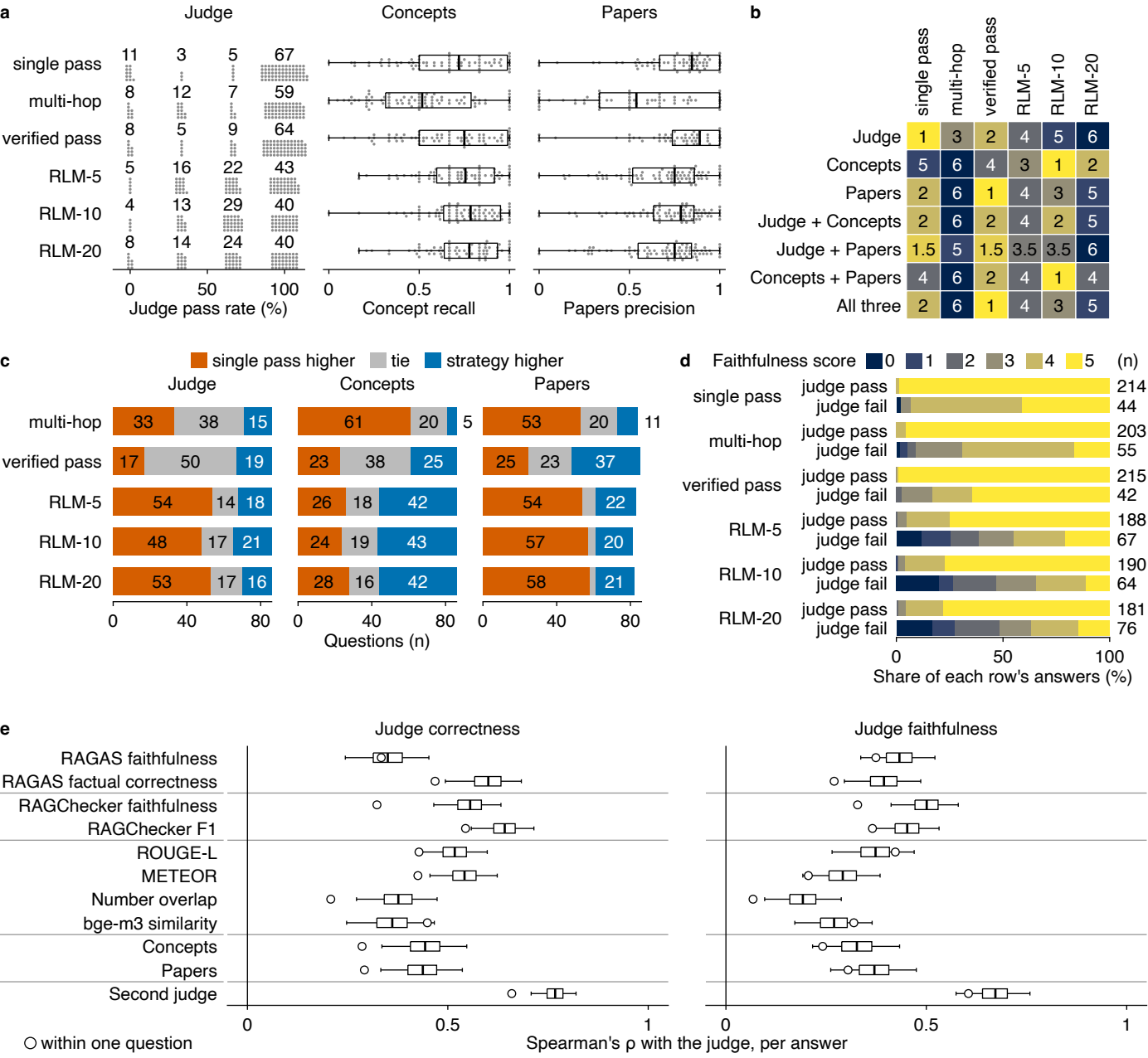

a
Judge
Concepts
Papers
single pass
multi-hop
verified pass
RLM-5
RLM-10
RLM-20
11 3 5 67
8 12 7 59
8 5 9 64
5 16 22 43
4 13 29 40
8 14 24 40
0 50 100
Judge pass rate (%)
0 0.5 1
Concept recall
0 0.5 1
Papers precision
b
single pass
multi-hop
verified pass
RLM-5
RLM-10
RLM-20
Judge 1 3 2 4 5 6
Concepts 5 6 4 3 1 2
Papers 2 6 1 4 3 5
Judge + Concepts 2 6 2 4 2 5
Judge + Papers 1.5 5 1.5 3.5 3.5 6
Concepts + Papers 4 6 2 4 1 4
All three 2 6 1 4 3 5
c
single pass higher
tie
strategy higher
Judge
Concepts
Papers
multi-hop 33 38 15 61 20 5 53 20 11
verified pass 17 50 19 23 38 25 25 23 37
RLM-5 54 14 18 26 18 42 54 22
RLM-10 48 17 21 24 19 43 57 20
RLM-20 53 17 16 28 16 42 58 21
0 40 80
Questions (n)
d
Faithfulness score
0
1
2
3
4
5
(n)
single pass
judge pass 214
judge fail 44
multi-hop
judge pass 203
judge fail 55
verified pass
judge pass 215
judge fail 42
RLM-5
judge pass 188
judge fail 67
RLM-10
judge pass 190
judge fail 64
RLM-20
judge pass 181
judge fail 76
0 50 100
Share of each row's answers (%)
e
Judge correctness
Judge faithfulness
RAGAS faithfulness
RAGAS factual correctness
RAGChecker faithfulness
RAGChecker F1
ROUGE-L
METEOR
Number overlap
bge-m3 similarity
Concepts
Papers
Second judge
0 0.5 1
0 0.5 1
○ within one question
Spearman's ρ with the judge, per answer